



How “Extreme” is Extreme Pain?

Patient Identity and Intensifier Usage Affects Physicians’ Perceptions of Pain

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How does a patient’s race, gender, and intensifier usage impact physicians’ perceptions of their pain?

Linguistic bias is the negative evaluation of speakers based on their voice, dialect, or language choices, and can lead to discrimination¹



Speaking in a different dialect
→ less housing callbacks¹



Speaking with an accent
→ less likely to be hired^{2, 3}



Speaking with an accent
→ less trustworthy and credible^{4, 5}

Physicians have negative perceptions of a patient (PT) based on race and gender⁶⁻⁸

Negative Perceptions of PTs

Black (vs. White) PTs

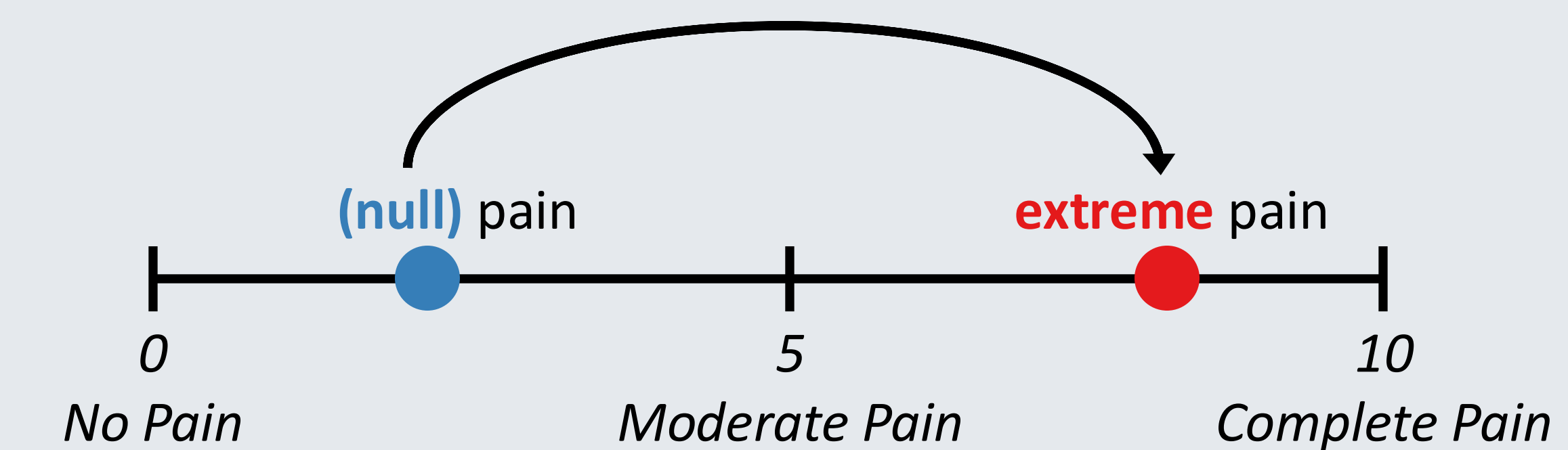
- Overly high pain tolerance⁶
- Understate pain⁶
- Distrusted pain reports⁷

Women (vs. Men) PTs

- Low pain tolerance^{6, 8}
- Exaggerate pain^{6, 8}
- Distrusted pain reports⁸

Patients report **emphasizing symptoms** to be heard/taken seriously⁹

Intensifiers, like 'extreme', boost the meaning of a word or phrase^{10, 11}



54 physicians read 8 hypothetical patients’ symptom reports and were asked to rate “How much pain do you believe the patient is in?” on an 11-point Likert scale

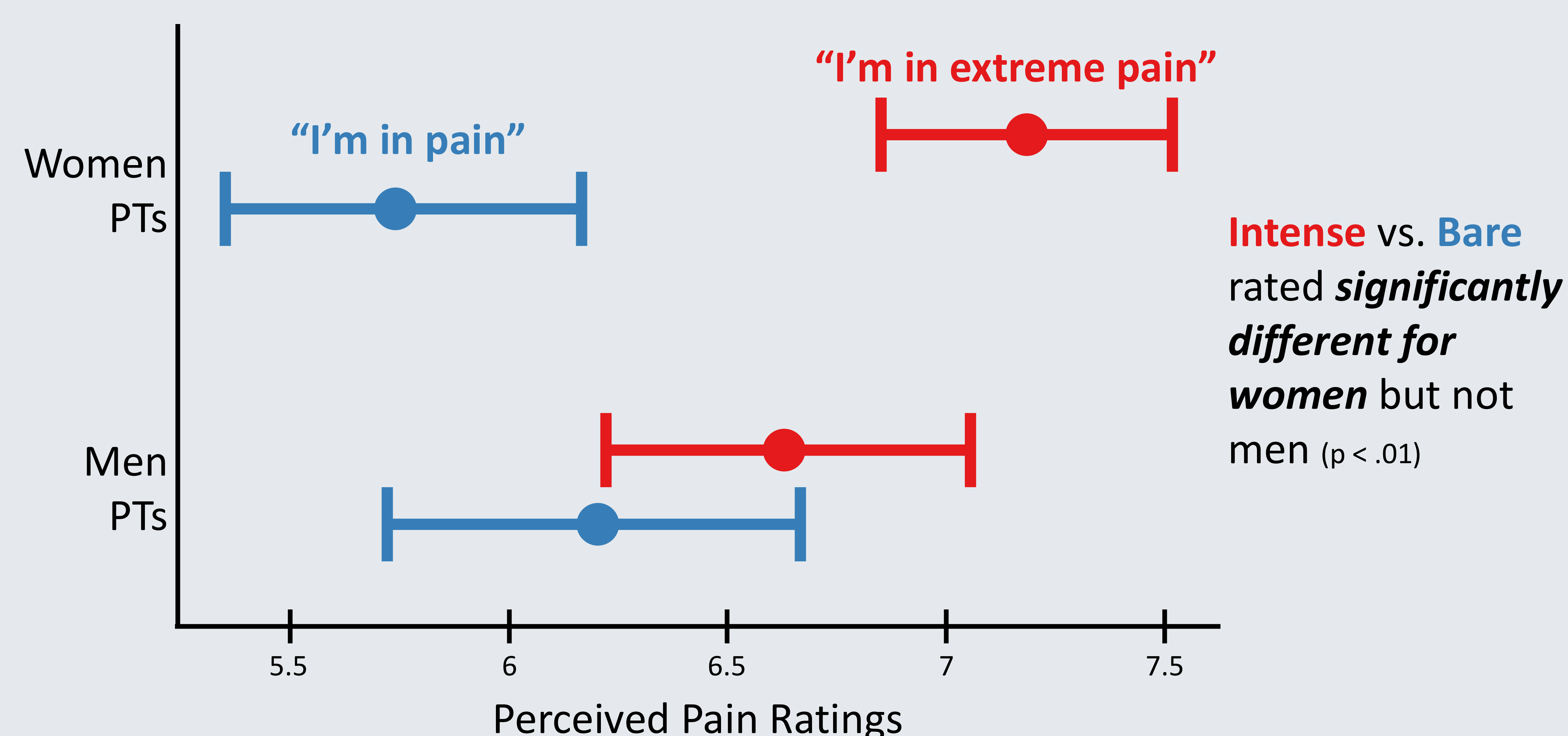
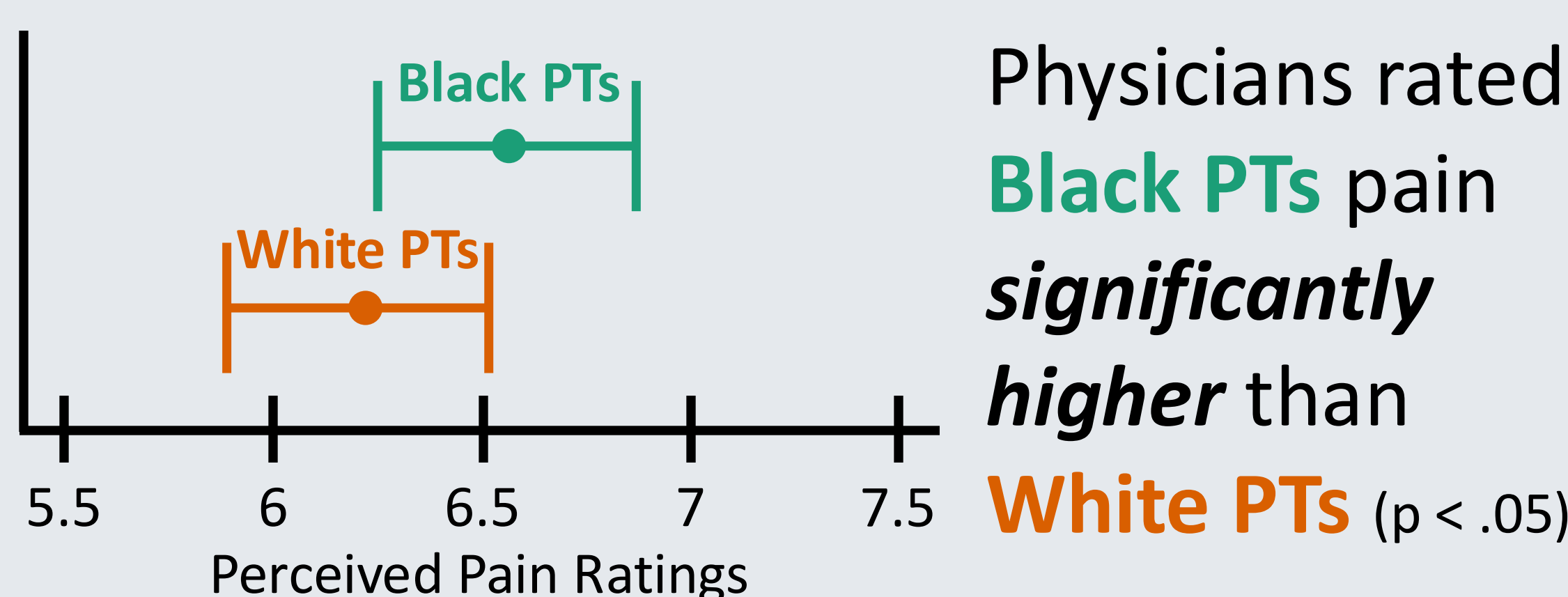
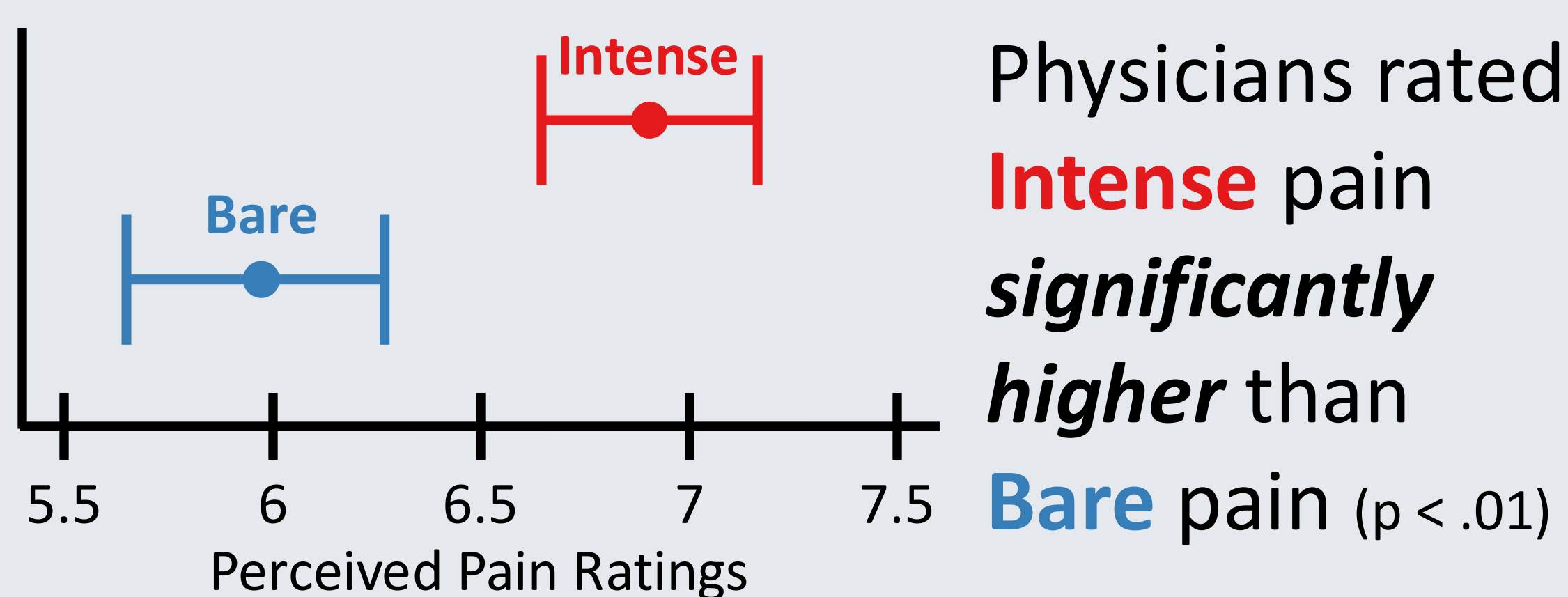
4 critical reports manipulated:

Patient Type	X	Intensity
Black Man		Bare
Black Woman		Bare
White Man		Intense
White Woman		Intense

A 56-year-old **Black woman** comes into your office. When you ask, “What brings you in today?”, the patient responds with:

I’ve been having **extreme** pain in my lower back and down both of my legs for several months. My left side hurts more, and I also sometimes have **severe** tingling and numbness in my thighs and legs, which makes it hard to walk comfortably. It takes **intense** effort to stand and lift items. ...

Intensifier interpretation *is* influenced by the speaker’s social characteristics



Physicians rated pain differently based on Intensity *and* Gender, supporting a **Socio-Semantic Hypothesis** (scan below)

Intensifier interpretation is mediated by social information (gender), providing new evidence for theories integrating linguistic meaning and social meaning

Linguistic variables are another factor in assessing biased medical outcomes

